Exercises 3.3 1, 3, 5, 7, 9, 11, 13, 15, 17, 18, 19, 20, 23, 33, 35, 39, 41, 43, 45, 51, 57 (*Hint*: $s = r\theta$, draw the bisector of the angle θ and find $\sin(\frac{\theta}{2})$).

1. $f(x) = 5\sin x - 2\cos x$

$f'(x) = 5\cos x + 2\sin x$

3. $y = x^2 + \cot x$

$y' = 2x - \csc^2 x$

5. $h(\theta) = \theta^2 \sin\theta$

$h'(\theta) = \theta\cos\theta + 2\theta\sin\theta$

$(1 + \tan^2\theta) = \sec^2\theta$

7. $y = \sec\theta\tan\theta$

$y' = \sec^2\theta + \sec\theta\tan^2\theta$

9. $f(\theta) = (\theta - \cos\theta)\sin\theta$

$f'(\theta) = (1 + \sin\theta)\sin\theta + (\theta - \cos\theta)\cos\theta$

11. $H(t) = \cos^2 t$

$H'(t) = +2\cos t \cdot \sin t$

13. $f(\theta) = \dfrac{\sin\theta}{1 + \cos\theta}$

$f'(\theta) = \dfrac{(1+\cos\theta)\cos\theta + \sin^2\theta}{(1+\cos\theta)^2} = \dfrac{\cos\theta + 1}{(1+\cos\theta)^2} = \dfrac{1}{1+\cos\theta}$

15. $y = \dfrac{x}{2 - \tan x}$

? 分子一定要化简吗

$y' = \dfrac{2 - \tan x + x \cdot \sec^2 x}{(2 - \tan x)^2}$

17. $f(w) = \dfrac{1 + \sec w}{1 - \sec w}$

$f'(w) = \dfrac{(1 - \sec w)\sec w \tan w + (1 + \sec w)\sec w \tan w}{(1 - \sec w)^2}$

$= \dfrac{2 \sec w \tan w}{(1 - \sec w)^2}$

18. $y = \dfrac{\sin t}{1 + \tan t}$

$y' = \dfrac{(1 + \tan t)\cos t - \sec^2 t \sin t}{(1 + \tan t)^2} = \dfrac{(1 + \frac{\sin t}{\cos t})\cos t - \frac{\sin t}{\cos^2 t}}{(1 + \tan t)^2} = \dfrac{\cos t + \sin t - \frac{\tan t}{\cos t}}{(1 + \tan t)^2}$

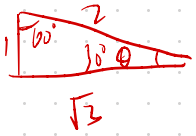
19. $y = \dfrac{t \sin t}{1 + t}$

$y' = \dfrac{(1+t)(\sin t + t\cos t) - t\sin t}{(1+t)^2}$ ✓ $\Rightarrow (t^2 + t)\cos t + \sin t$

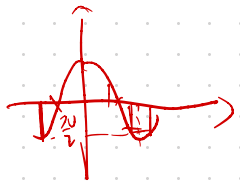
20. $g(\theta) = \dfrac{\theta}{\sec\theta + \tan\theta}$

$g'(\theta) = \dfrac{\sec\theta + \tan\theta - \theta(\sec\theta\tan\theta + \sec^2\theta)}{(\sec\theta + \tan\theta)^2}$ $\Rightarrow 1 - \theta\sec\theta$

33. a) $f'(x) = \sec x \tan x - 1$



35. $g'(\theta) = \dfrac{\theta\cos\theta - \sin\theta}{\theta^2}$



π/3 60°

120

39. $f(x) = x + 2\sin x$

$f'(x) = 1 + 2\cos x = 0$

$\cos x = -\frac{1}{2}$, when $x \in \frac{2\pi}{3} + 2\pi n$, $\frac{4\pi}{3} + 2\pi n$

$\frac{2\pi}{3}$

41. velocity: $x'(t) = 8\cos t$

acceleration: $x''(t) = -8\sin t$

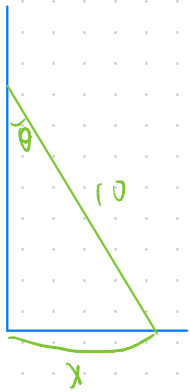
$\frac{4\pi}{3} = 240°$

b) position: $x(\frac{2\pi}{3}) = 8\sin(\frac{2\pi}{3}) = 6.93$

v: $x'(\frac{2\pi}{3}) = 8\cos(\frac{2\pi}{3}) = -4$

a: $x''(\frac{2\pi}{3}) = -8\sin(\frac{2\pi}{3}) = -6.93$

43. $\sin\theta = \frac{x}{10} \Rightarrow x = 10\sin\theta$



$x' = 10\cos\theta$

$x'(\frac{\pi}{3}) = 10\cos(\frac{\pi}{3})$

$= 5$

(12) $\lim_{x\to 0} \frac{\tan x}{x} = \lim_{x\to 0} \frac{\sin x}{x\cos x} = \lim_{x\to 0} \frac{\sin x}{x} \cdot \lim_{x\to 0} \frac{1}{\cos x}$

$= 1 \times 2 = 2$

不等関数

$\lim_{\theta\to 0} \frac{\sin\theta}{\theta} = 1$

$\lim_{\theta\to 0} \frac{1-\cos\theta}{\theta} = 0$

(13) $\lim_{\theta\to 0} \frac{1-\cos\theta}{2\theta^2} = \lim_{\theta\to 0} \frac{-(1-\cos\theta)}{\theta} \cdot \lim_{\theta\to 0} \frac{1}{2\theta}$ DNE

$= 0$

$\lim_{\theta\to 0} \frac{1-\cos\theta}{2\theta^2} \cdot \frac{1+\cos\theta}{1+\cos\theta} = \lim_{\theta\to 0} \frac{1-\cos^2\theta}{2\theta^2(1+\cos\theta)}$

$= \lim_{\theta\to 0} \frac{\sin^2\theta}{2\theta^2(1+\cos\theta)}$

$= \frac{1}{2} \lim_{\theta\to 0} \frac{\sin\theta}{\theta} \cdot \frac{\sin\theta}{\theta} \cdot \frac{1}{1+\cos\theta}$

$= \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$